

Identification of fossil pollen

Ingrid Romero

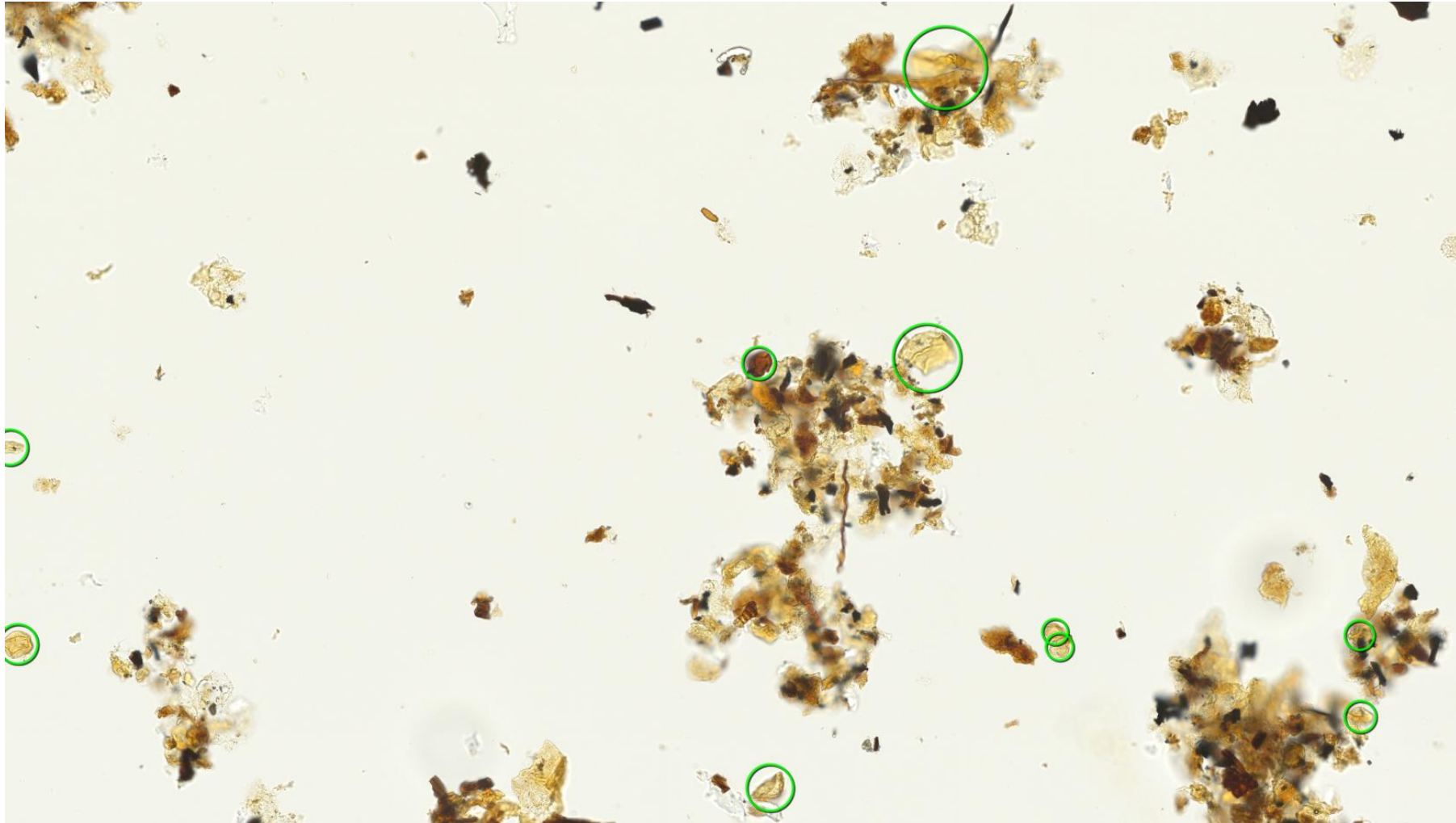


What motivates this project?

- We are paleontologists interested in past climates – particularly in warm periods when there were no polar ice caps. We want to understand those past warm climates better because they are key to making good predictions about future climate change under high CO₂.
- Plants are very sensitive to climate, so plant FOSSILS are good for describing the distribution of past climates.
- The most abundant and widespread plant fossils are...believe it or not...POLLEN!

- The fossils are 50-60 million years old – so after the dinosaurs and long before humans.
- The data you will be working with are images of fossil pollen that come from a very large collection – tens of thousands of slides with as many as 10,000 grains on just one slide.
- The slides come from all over North America.
- With such a large and diverse collection, we need AI help to identify the fossils.

Here is what a slide image looks like



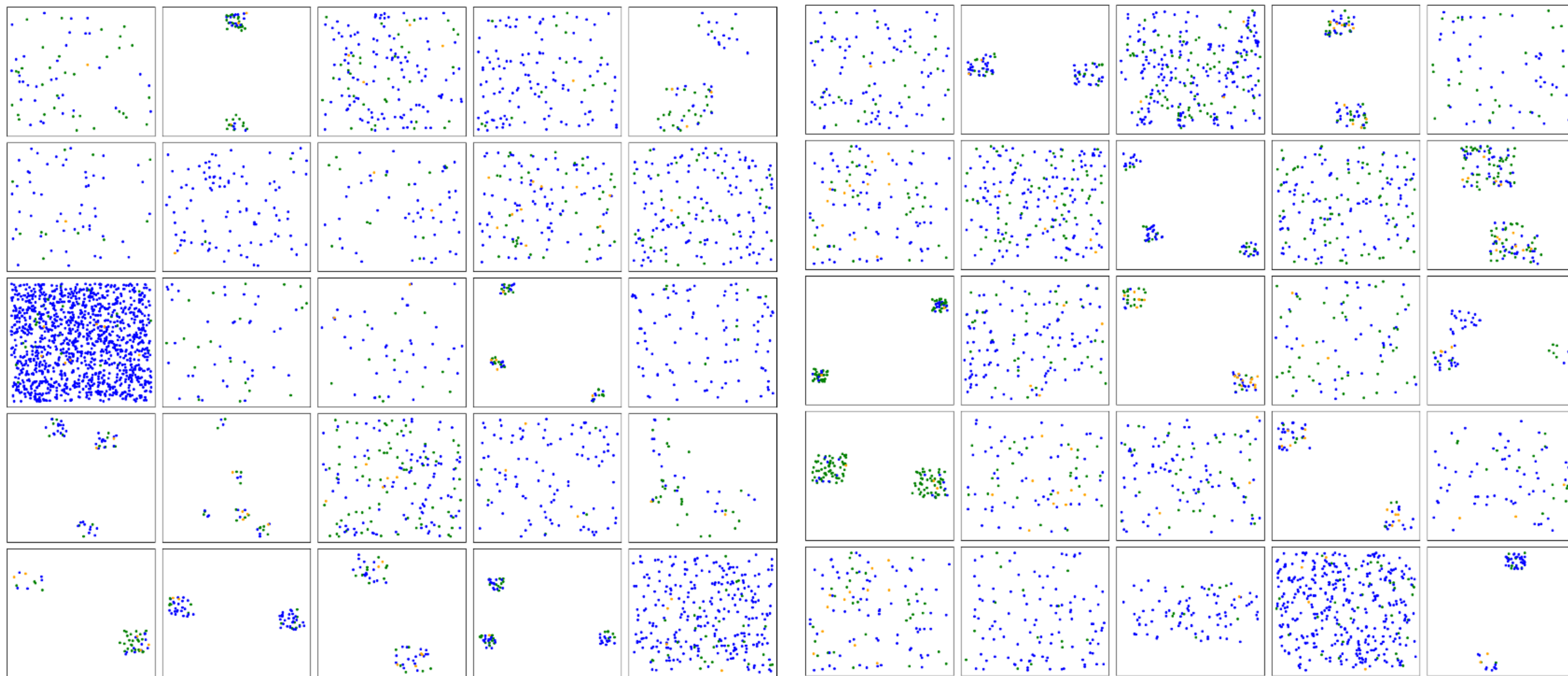
You might notice that there are many bits on this slide that have NOT been identified.

These are fragments of plants, fungi and other organic matter.

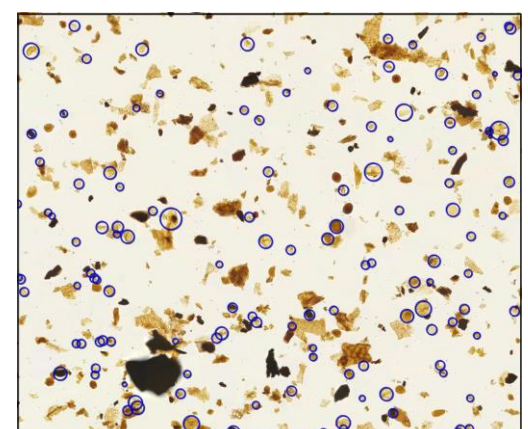
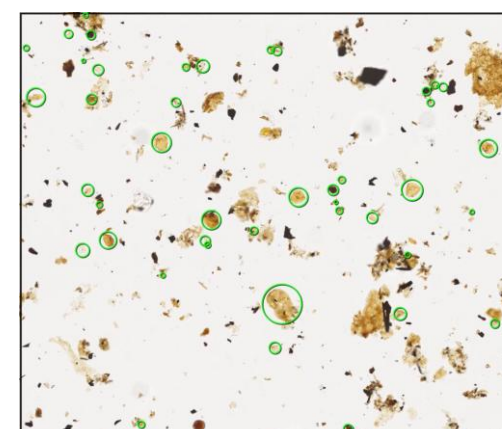
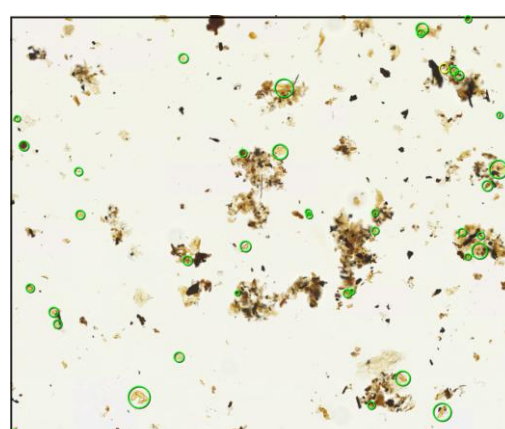
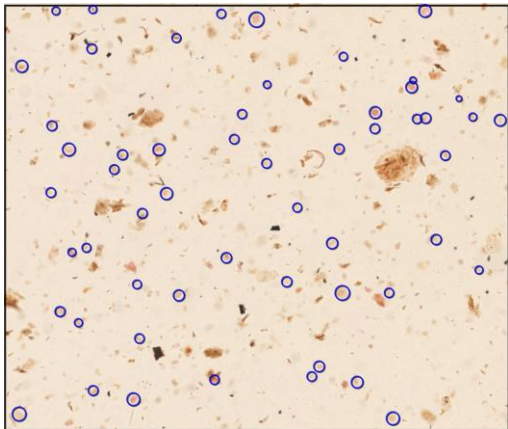
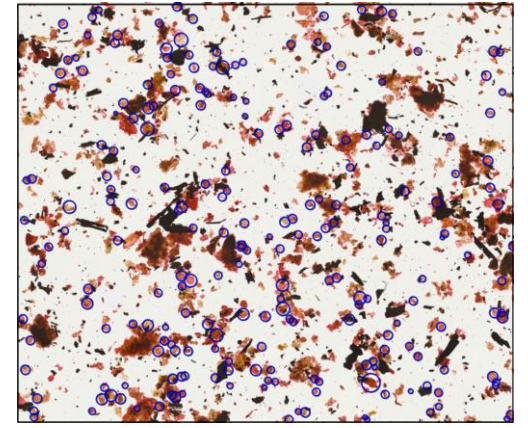
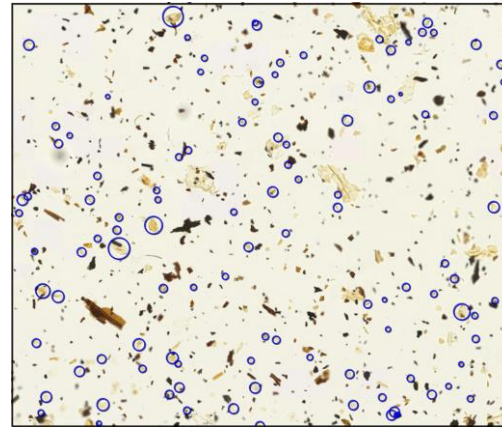
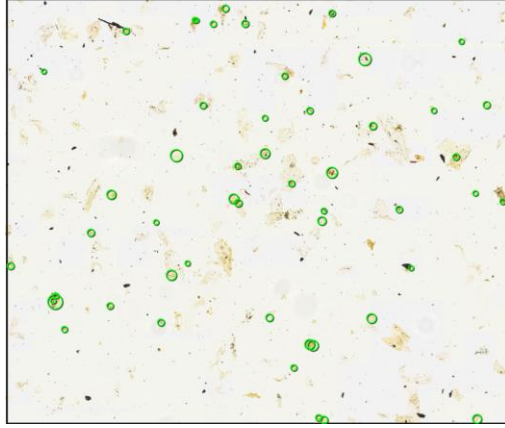
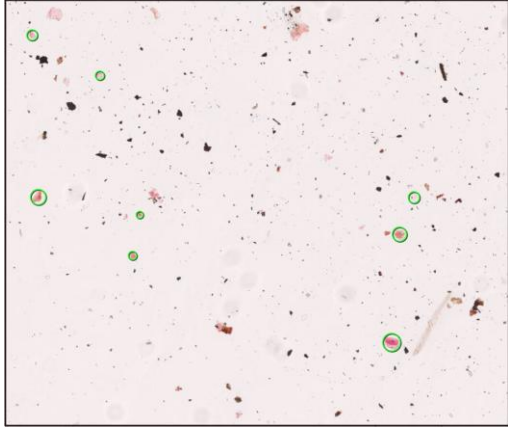
They are not any of the seven types we have defined

Diversity of the pollen slides

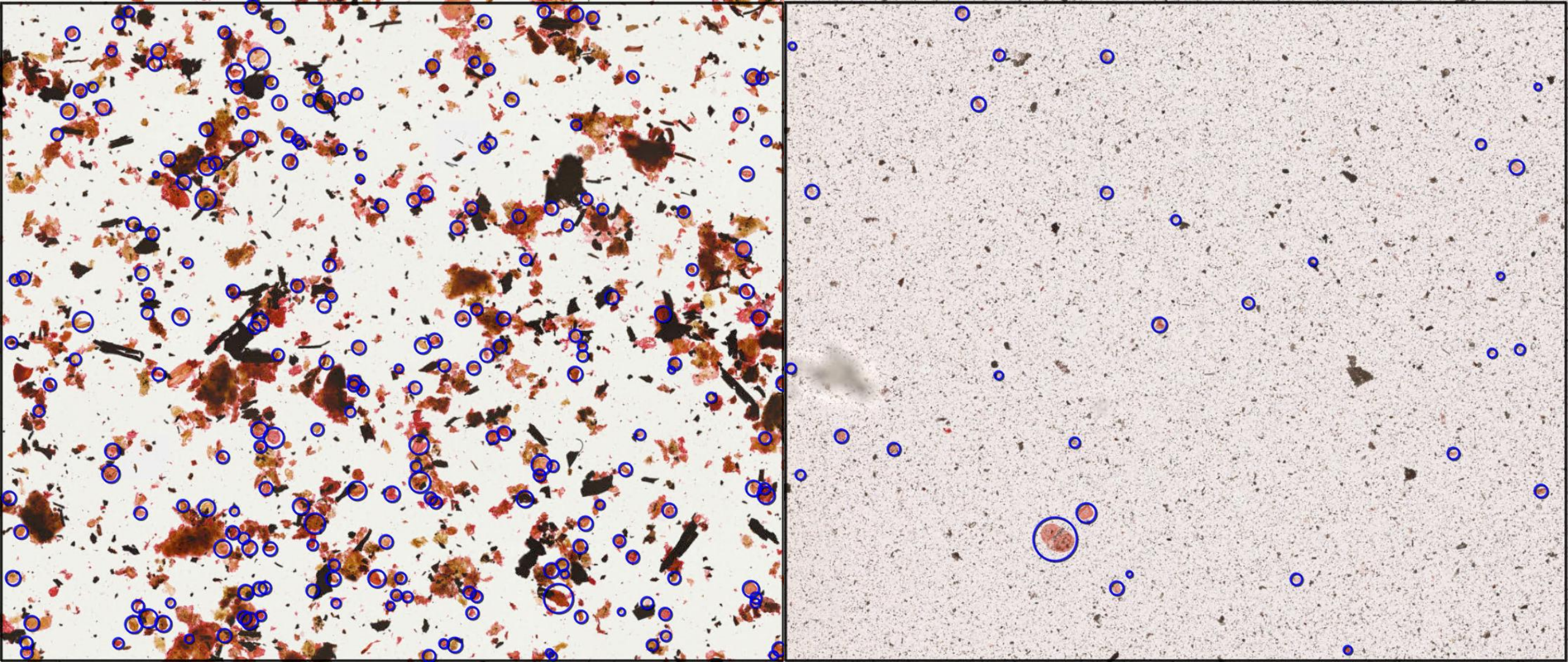
(each blue or green dot is a human-identified region of interest)



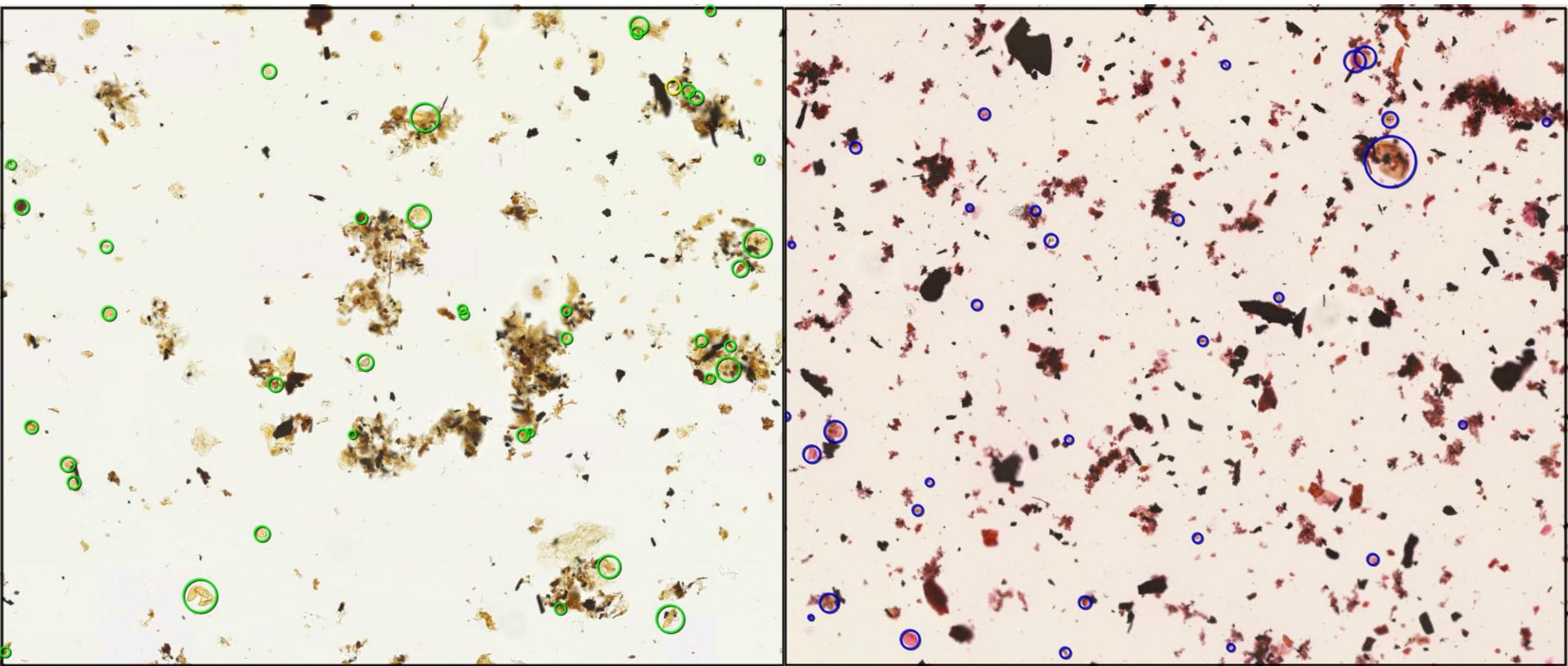
The slides are visually diverse – 100s of different types of pollen.
Here is a sample of the diversity of slide appearance across the data set.



The slides are visually diverse – different density of pollen and non-pollen organic bits, different types of non-pollen organic bits



The slides are visually diverse – different mounting media, different staining to enhance the pollen





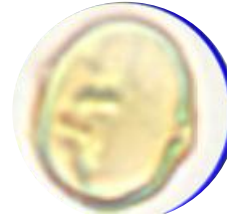
Platycarya platycarioides



Cicatricosisporites dorogensis



Momipites flexus



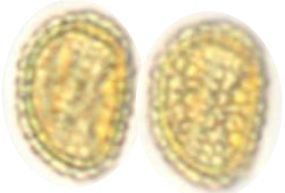
Tetracolporopollenites sp.
(Sapotaceae)



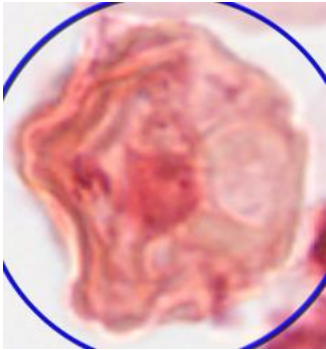
Caryapollenites veripites



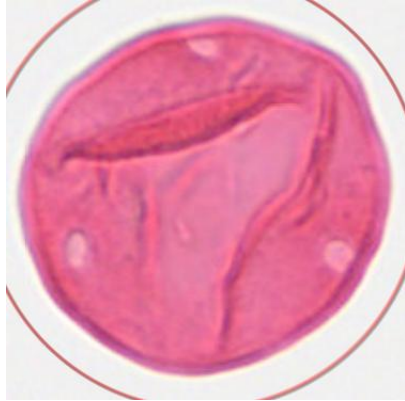
Caryapollenites wodehousei



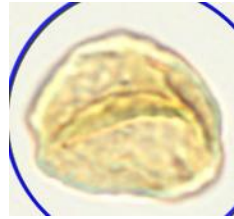
Caprifoliipites sp.



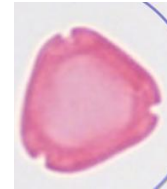
Alnipollenites scoticus



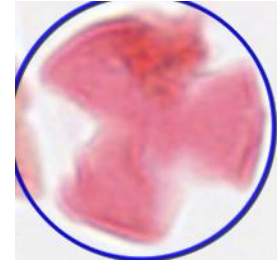
Carya sp.



Ulmipollenites sp.



Momipites
wyomingensis



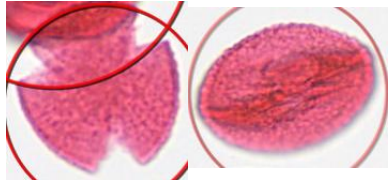
Psilatricolpites sp.



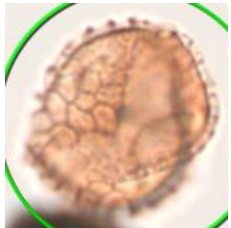
Taxodium sp.



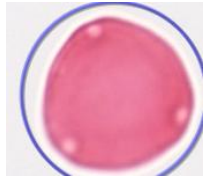
Quercus sp.



Verrutricolpites sp.



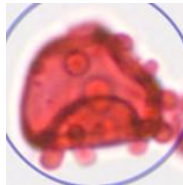
Amanoa sp.



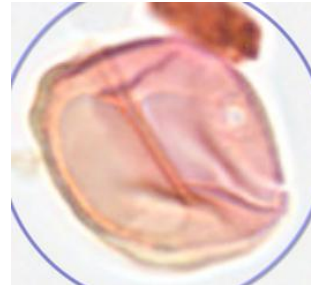
Caryapollenites imparalis



Polyatriopollenites vermontensis



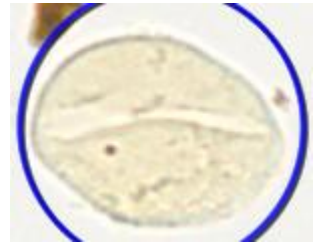
Pistillipollenites mcgregorii



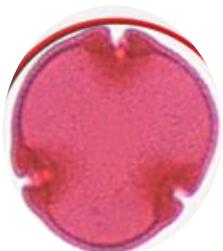
Juglans nigripites



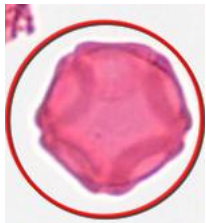
Bissacate



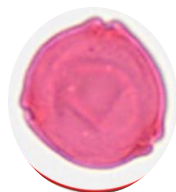
Psilamonocolpites sp.



Bombacacidites sp.



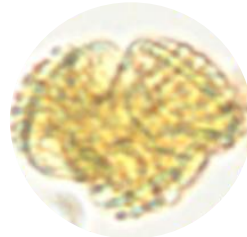
Alnipollenites verus



Betulaceae



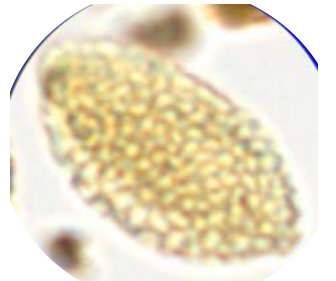
Ericipites sp.



Illexpollenites sp.

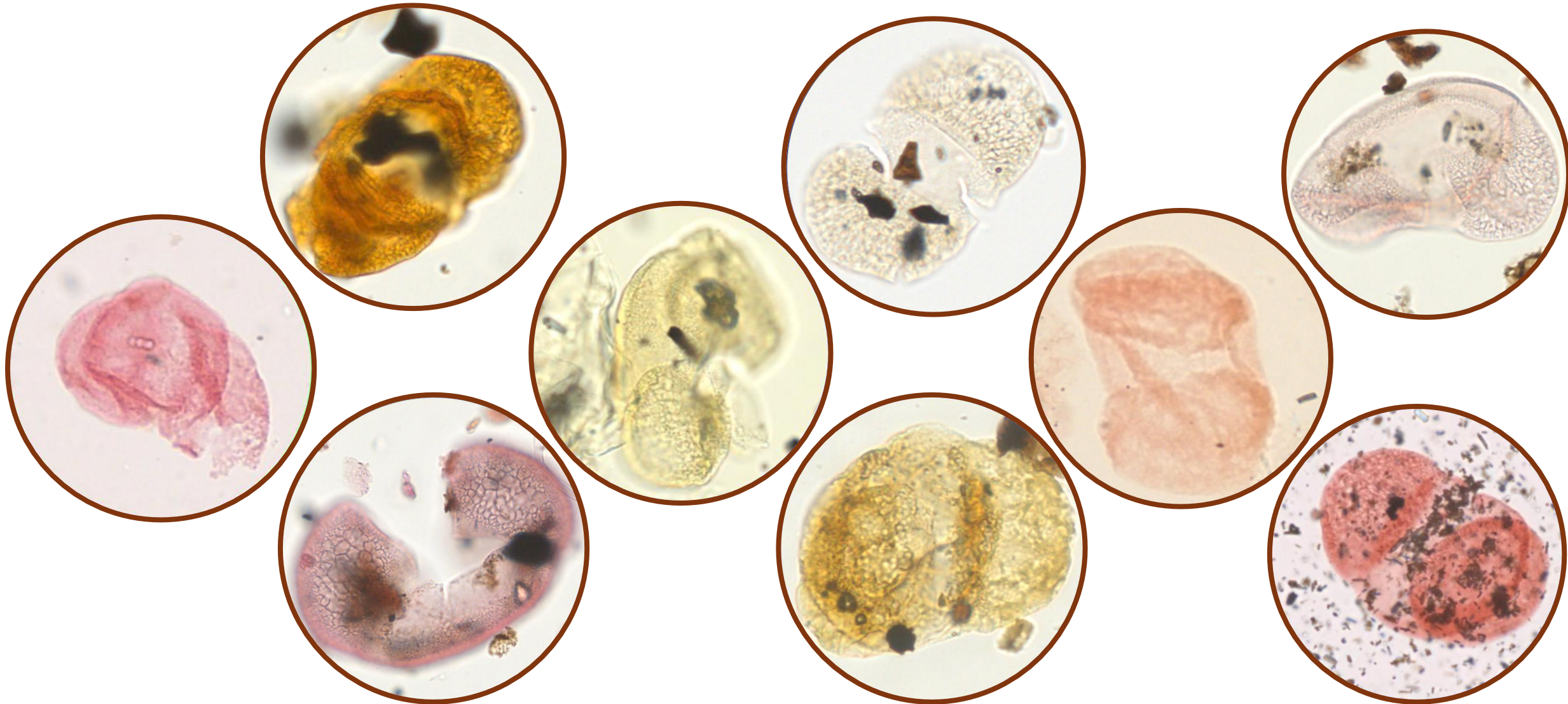


Laevigatosporites sp.

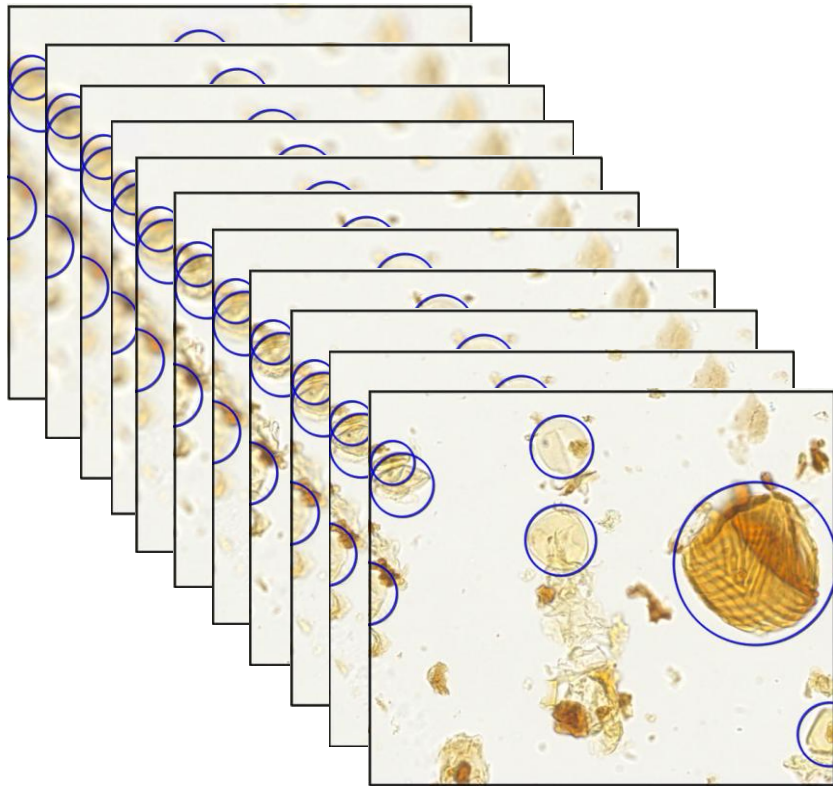


Arecipites sp.

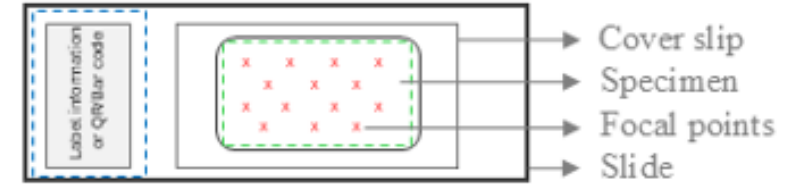
Examples of bisaccate pollen showing variations of this pollen type based on preservation (color, breaks), view position, and stain method.



Pollen is 3D, and we have multiple planes of focus in each image and the information from those multiple planes is critical for identifying the pollen.



Focal points



Z-Stack



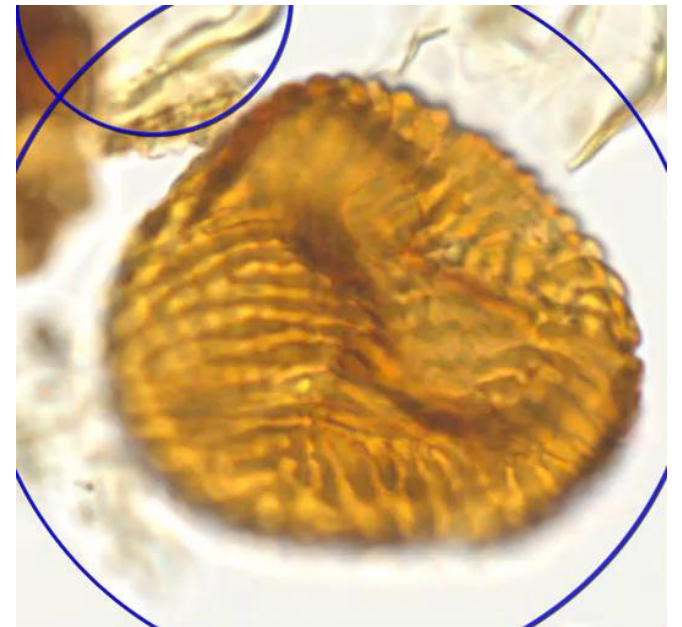
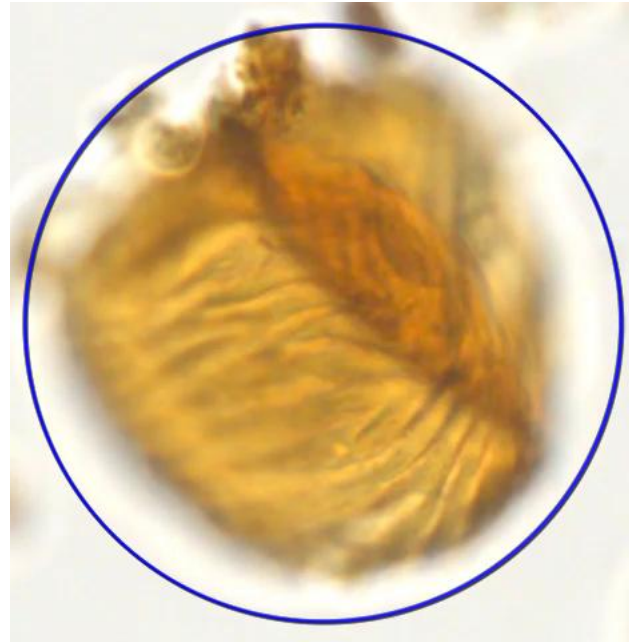
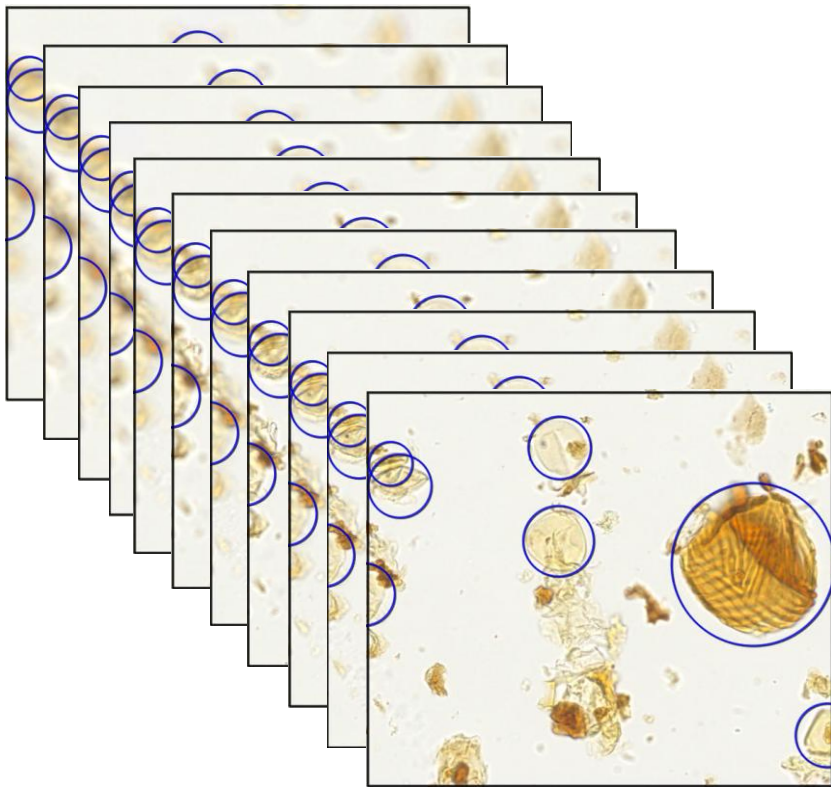
Focal points in the scanned z-space



Conventions

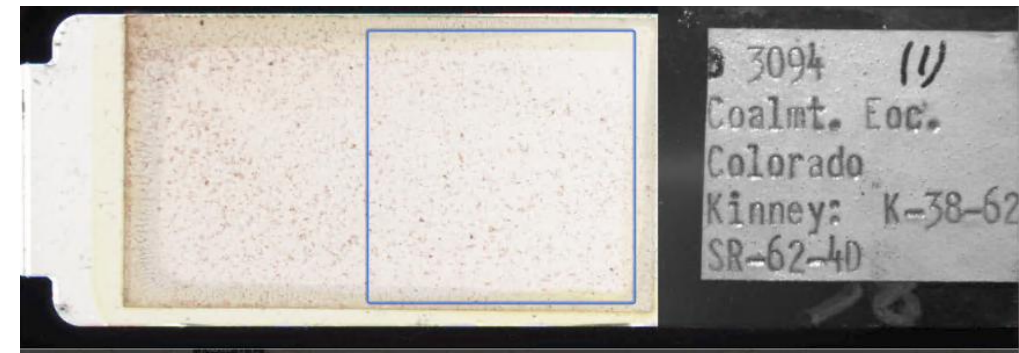
- - - Label area
- - - Scan area
- x Focal point

Pollen is 3D, and we have multiple planes of focus in each image and the information from those multiple planes is critical for identifying the pollen.



Each image is a scanned area of 400mm^2 from each slide.

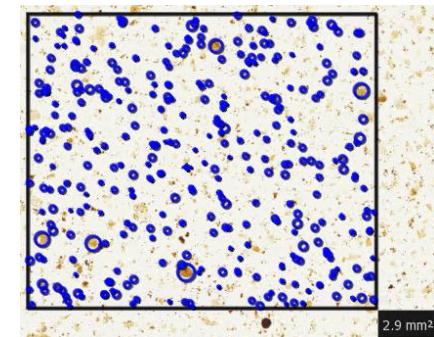
From there, a smaller area of 2.9mm^2 was selected to annotate all pollen found in it. This area is the one to use for the training set

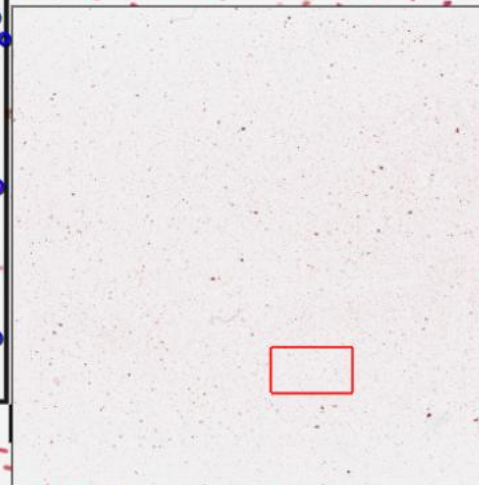
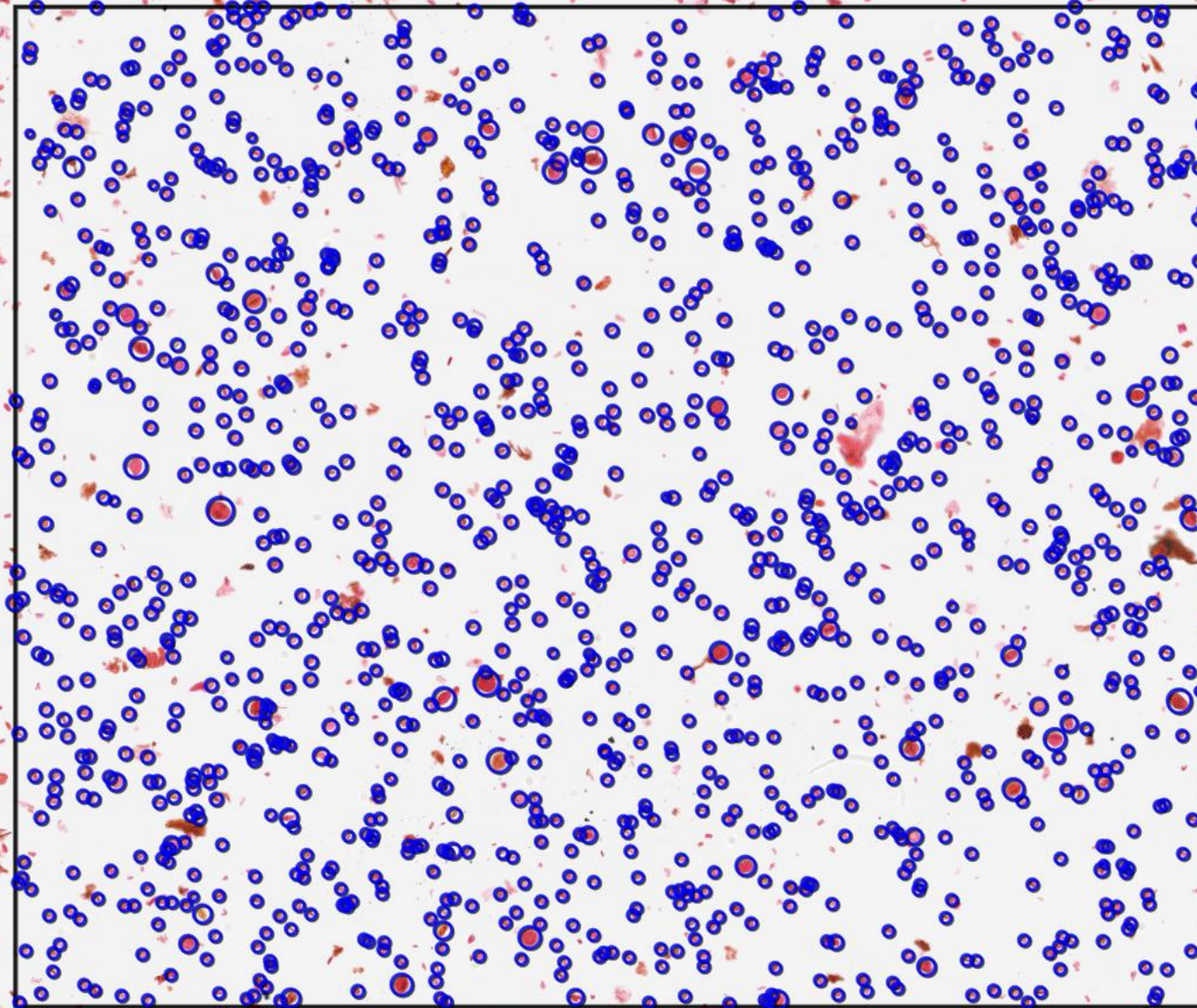
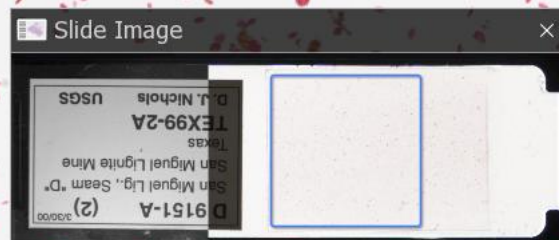


Scanned area from
slide
(20 x 20 mm)



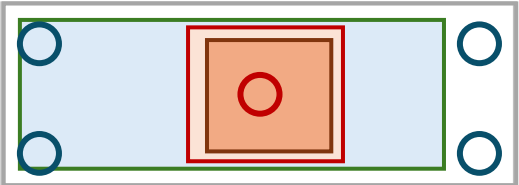
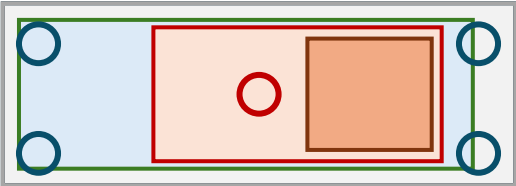
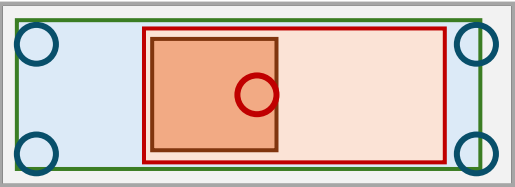
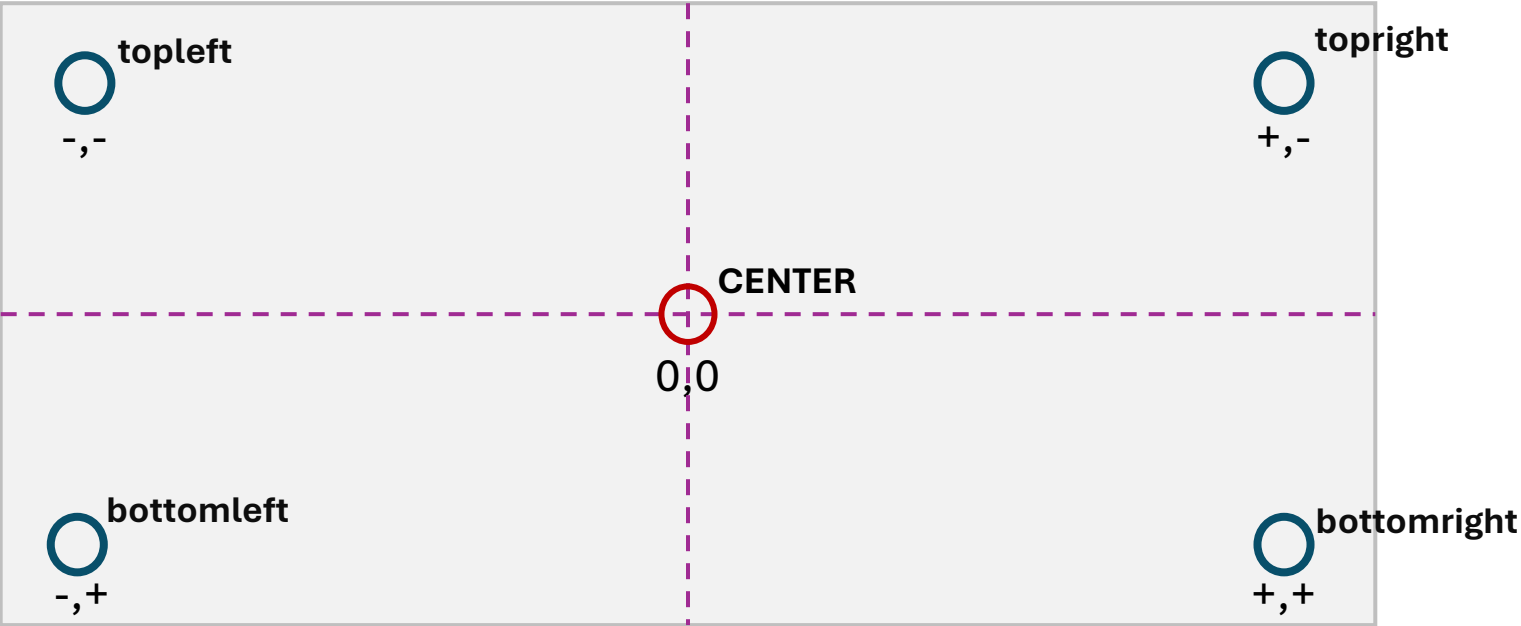
Selected area for
training
(1.7x1.7mm)







Graphic illustration of the coordinate plane of the NanoZoomer



Legend

Nanozoomer area

Microscope slide

Cover slip

Scanned area

+,+

 Points in coordinate plane

Center of the NZ area

Other regions of interest



topleft



topright



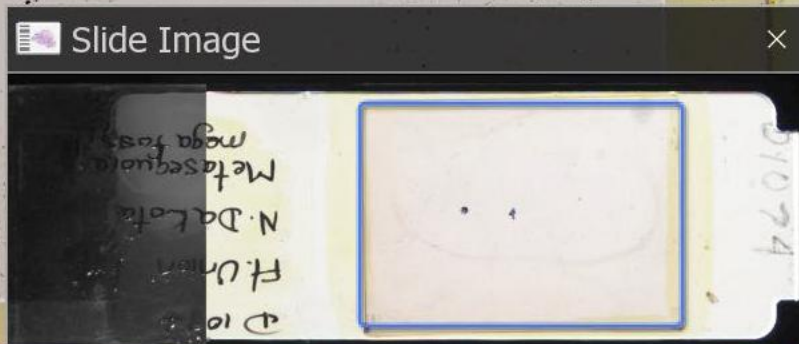
CENTER 0-0

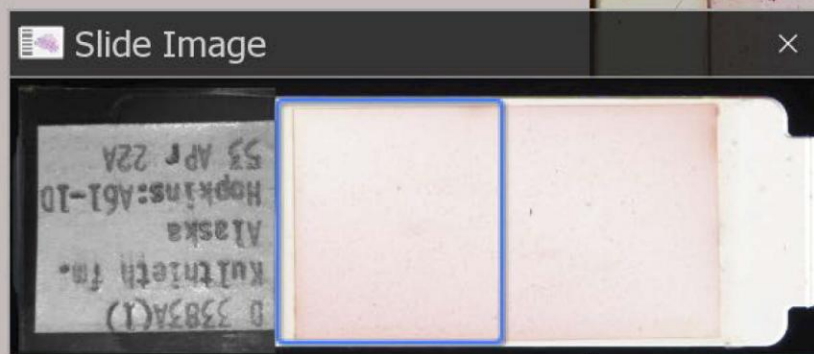
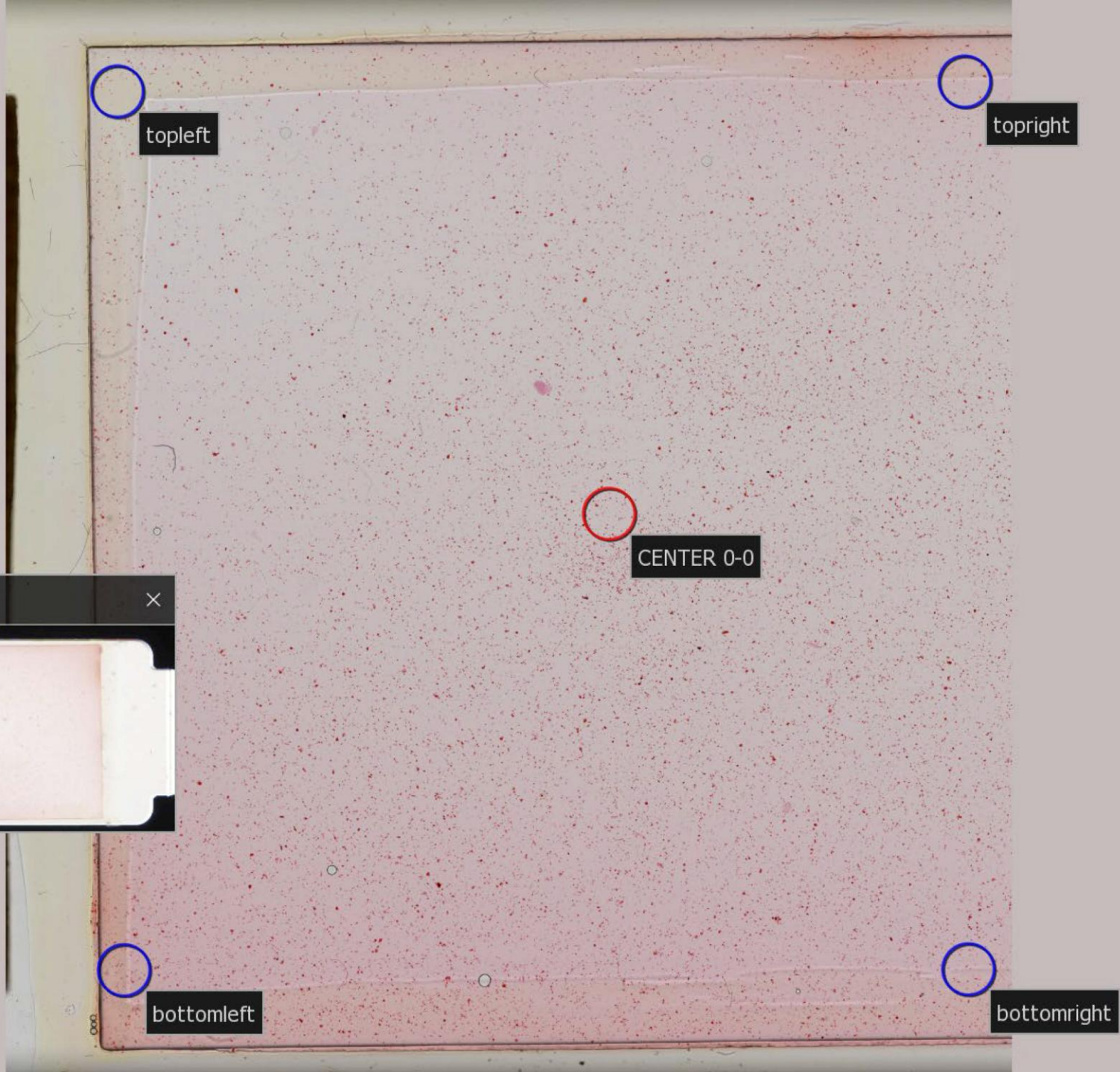


bottomleft



bottomright







opleft



topright



CENTER 0-0



bottomleft



bottomright

